TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	76.0 ± 3.58
Correlation Stability	85.0 ± 3.00
Distribution Consistency	85.8 ± 3.19
Entropy Preservation	87.3 ± 1.42
Mean Z Score Drift	0.2 ± 0.03
Categorical Fidelity	81.5 ± 0.50
Privacy Risk	27.5 ± 1.75
Data Density	99.9 ± 0.14
Coverage Score	29.7 ± 4.45
Feature Accuracy	76.7 ± 2.10
Novelty Score	0.6 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	10%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	42.502	-2.244	5.0%	OK
TB	3.299	1.749	-1.550	47.0%	Drift
DB	1.486	0.700	-0.786	52.9%	Drift
Alkphos	290.576	213.015	-77.561	26.7%	Drift
Sgpt	80.714	38.805	-41.909	51.9%	Drift
Sgot	109.911	45.245	-64.666	58.8%	Drift
TP	6.483	6.746	+0.263	4.1%	OK
ALB	3.142	3.361	+0.219	7.0%	OK
A/G Ratio	0.947	0.905	-0.042	4.5%	OK
Selector	1.286	1.307	+0.020	1.6%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	76.0 ± 3.58
Correlation Stability (*2)	85.0 ± 3.00
Distribution Consistency (*3)	85.8 ± 3.19
Entropy Preservation (*4)	87.3 ± 1.42
Mean Z Score Drift (*5)	0.2 ± 0.03
Categorical Fidelity (*6)	81.5 ± 0.50
Privacy Risk (*7)	27.5 ± 1.75
Data Density (*8)	99.9 ± 0.14
Coverage Score (*9)	29.7 ± 4.45

Feature Accuracy (*10)	76.7 ± 2.10
Novelty Score (*11)	0.6 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6589743589743591	0.5521367521367522	0.6564102564102564	-0.00256410256410255	
F1	0.7616166093560027	0.6494631727391557	0.7582909047304992	-0.00332570462550353 4	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6376068376068377	0.6923076923076923	0.01794871794871794	
F1	0.7821343746683586	0.7465765732345057	0.8002454398124316	0.01811106514407304 8	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7111111111111111	0.5974358974358974	0.6888888888888889	-0.02222222222222222	
F1	0.8176075617756519	0.7002719474466051	0.7996554310347644	-0.01795213074088739 7	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7085470085470086	0.6384615384615384	0.7145299145299144	0.00598290598290598 4	
F1	0.8124574819778058	0.7498761981044784	0.8159994873944705	0.00354200541666452 9	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6068376068376069	0.7051282051282051	0.0025641025641025663	
F1	0.7997903571549948	0.720172587202897	0.7998454777072911	5.512055229626167e-05	

Model: KNN

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6649572649572655	0.6854700854700855	-0.00085470085470084	
F1	0.7971299319768493	0.7676355256282019	0.7920408278328198	-0.00508910414402964	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.617948717948718	0.5786324786324786	0.01965811965811968	*
F1	0.5587432119380917	0.68797390703879	0.5908209639420205	0.03207775200392895	*

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6811965811965812	0.6034188034188034	0.6982905982905983	0.01709401709401708	
F1	0.7879854100702532	0.711809775812236	0.7988434513538926	0.01085804128363935 2	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6794871794871795	0.7136752136752136	0.00683760683760683 6	
F1	0.8279091624529537	0.7875198137180976	0.8263757497140976	-0.00153341273885611 02	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.611965811965812	0.6957264957264957	-0.01025641025641025 3	
F1	0.8128002697439015	0.726579386003371	0.802510379411031	-0.01028989033287061	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 10% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3
2	39	Male	0.6	0.2	188	28	43	8.1	3.3	0.6
3	40	Male	1.9	1.0	231	16	55	4.3	1.6	0.6
4	40	Male	1.2	0.6	204	23	27	7.6	4.0	1.1
5	45	Female	0.9	0.3	189	23	33	6.6	3.9	nan
6	43	Male	0.8	0.2	192	29	20	6.0	2.9	0.9
7	25	Female	0.4	0.1	150	22	30	6.5	3.8	1.2
8	37	Male	0.9	0.3	200	30	35	7.2	4.0	1.5
9	52	Female	1.2	0.4	180	25	28	6.8	3.5	1.0
10	45	Male	0.7	0.2	210	40	50	7.5	4.2	1.3
11	29	Female	0.6	0.2	160	20	25	6.0	3.0	1.1
12	34	Male	1.5	0.5	220	35	40	7.0	3.9	1.4
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 10% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	40	Male	14.5	6.4	358	50	75	5.7	2.1	0.5
2	27	Male	1.3	0.6	106	25	54	8.5	4.8	nan
3	75	Male	1.4	0.4	215	50	30	5.9	2.6	0.7
4	36	Male	0.9	0.1	486	25	34	5.9	2.8	0.9
5	42	Male	0.8	0.2	158	27	23	6.7	3.1	0.8
6	48	Male	0.7	0.2	326	29	17	8.7	5.5	1.7
7	34	Male	1.2	0.5	210	45	60	6.5	3.0	0.8
8	29	Female	0.9	0.3	180	30	40	7.2	3.5	1.0
9	55	Male	3.5	1.2	400	60	80	5.8	2.7	0.6
10	47	Female	2.1	0.8	250	50	70	6.2	3.1	0.9
11	38	Male	1.5	0.6	300	55	65	7.0	4.0	1.2
12	60	Female	0.8	0.2	220	35	45	6.9	3.4	1.0
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 10% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Male	20.2	11.7	188	47	32	5.4	2.3	0.7
2	21	Male	1.0	0.3	142	27	21	6.4	3.5	1.2
3	22	Male	0.6	0.2	202	78	41	8.0	3.9	0.9
4	33	Male	0.7	0.2	256	21	30	8.5	3.9	0.8
5	65	Female	0.7	0.2	406	24	45	7.2	3.5	0.9
6	45	Male	2.9	1.4	210	74	68	7.2	3.6	1.0
7	34	Female	1.5	0.5	210	45	30	6.0	3.0	1.0
8	29	Male	0.9	0.3	180	35	25	7.2	3.8	1.2
9	50	Female	2.0	0.8	250	60	40	6.5	3.2	1.1
10	41	Male	0.7	0.2	190	50	35	7.0	3.6	1.0
11	55	Female	3.1	1.5	300	70	50	6.8	3.4	0.9
12	62	Male	1.8	0.6	220	40	30	5.9	2.8	1.1
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 10% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Female	0.9	0.3	189	23	33	6.6	3.9	nan
2	26	Male	0.6	0.2	120	45	51	7.9	4.0	1.0
3	31	Male	0.8	0.2	198	43	31	7.3	4.0	1.2
4	55	Female	8.2	3.9	1350	52	65	6.7	2.9	0.7
5	26	Male	1.3	0.4	173	38	62	8.0	4.0	1.0
6	32	Male	12.7	8.4	190	28	47	5.4	2.6	0.9
7	29	Female	0.7	0.2	150	30	40	6.5	3.5	1.0
8	55	Male	3.5	1.5	200	60	70	7.0	4.0	0.8
9	42	Female	1.2	0.4	180	50	55	6.8	3.9	1.1
10	34	Male	0.9	0.3	220	45	65	7.5	4.2	0.9
11	60	Female	2.5	1.0	300	80	90	6.0	3.0	1.2
12	27	Male	0.4	0.1	130	25	35	5.5	2.8	1.3
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 10% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	23	Male	1.0	0.3	212	41	80	6.2	3.1	1.0
2	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
3	58	Male	1.0	0.5	158	37	43	7.2	3.6	1.0
4	45	Male	1.1	0.4	92	91	188	7.2	3.8	1.11
5	54	Female	22.6	11.4	558	30	37	7.8	3.4	0.8
6	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
7	29	Female	0.9	0.2	150	30	45	6.5	3.0	0.8
8	34	Male	1.5	0.6	200	40	50	7.0	3.5	1.0
9	50	Female	0.8	0.1	180	25	30	6.2	3.1	0.9
10	39	Male	2.0	0.9	220	55	60	7.5	3.8	1.2
11	45	Female	1.1	0.4	210	35	28	6.8	3.4	1.0
12	60	Male	0.7	0.2	190	20	25	6.0	2.8	0.7
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 10% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	25	Male	0.8	0.1	130	23	42	8.0	4.0	1.0
2	58	Male	1.7	0.8	188	60	84	5.9	3.5	1.4
3	66	Male	1.0	0.3	190	30	54	5.3	2.1	0.6
4	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3
5	60	Male	2.1	1.0	191	114	247	4.0	1.6	0.6
6	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
7	34	Female	1.2	0.5	150	45	50	7.0	3.8	1.2
8	29	Male	0.9	0.2	200	30	40	6.5	3.2	1.0
9	55	Female	3.0	1.0	220	60	70	6.0	3.0	1.0
10	43	Male	1.5	0.6	180	50	60	6.8	3.4	1.1
11	37	Female	0.8	0.3	160	40	45	7.2	3.6	1.2
12	62	Male	2.5	1.2	240	70	80	5.5	2.9	0.9
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 10% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
2	31	Female	0.8	0.2	158	21	16	6.0	3.0	1.0
3	51	Male	0.8	0.2	175	48	22	8.1	4.6	1.3
4	38	Male	0.7	0.2	216	349	105	7.0	3.5	1.0
5	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
6	60	Male	0.8	0.2	215	24	17	6.3	3.0	0.9
7	34	Female	1.2	0.5	210	45	35	6.5	3.2	1.0
8	27	Male	0.9	0.3	180	30	25	7.0	3.8	1.1
9	45	Female	2.5	1.0	220	50	40	6.8	3.4	1.2
10	53	Male	3.0	1.5	250	60	45	7.2	3.5	1.3
11	39	Female	1.8	0.6	190	42	30	6.3	3.1	1.0
12	62	Male	2.2	0.7	300	70	50	7.5	3.9	1.4
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 10% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	31	Female	0.8	0.2	158	21	16	6.0	3.0	1.0
2	38	Female	2.6	1.2	410	59	57	5.6	3.0	0.8
3	32	Male	22.7	10.2	290	322	113	6.6	2.8	0.7
4	4	Male	0.9	0.2	348	30	34	8.0	4.0	1.0
5	66	Male	17.3	8.5	388	173	367	7.8	2.6	0.5
6	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
7	25	Female	1.0	0.3	150	20	15	6.5	3.2	1.1
8	42	Male	3.5	1.5	250	45	60	7.0	3.8	1.2
9	37	Female	0.9	0.4	200	30	25	6.8	3.5	1.0
10	29	Male	1.2	0.5	180	35	40	6.2	3.1	1.3
11	54	Female	2.1	0.7	220	50	55	7.5	4.0	1.25
12	31	Male	0.8	0.2	170	25	30	6.4	3.0	1.2
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 10% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	52	Male	0.9	0.2	156	35	44	4.9	2.9	1.4
2	35	Male	1.6	0.7	157	15	44	5.2	2.5	0.9
3	75	Male	10.6	5.0	562	37	29	5.1	1.8	0.5
4	38	Male	0.7	0.2	216	349	105	7.0	3.5	1.0
5	45	Male	3.2	1.4	512	50	58	6.0	2.7	0.8
6	70	Male	1.3	0.3	690	93	40	3.6	2.7	0.7
7	29	Female	0.5	0.1	150	20	30	6.5	3.2	1.2
8	34	Male	1.0	0.3	200	45	50	7.0	3.8	1.5
9	41	Female	0.8	0.2	180	25	40	6.8	3.5	1.3
10	55	Male	2.5	1.0	300	60	70	7.2	4.0	1.4
11	62	Female	3.0	1.5	250	55	65	6.9	3.9	1.6
12	48	Male	1.2	0.4	220	50	60	7.1	3.6	1.2
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 10% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	68	Female	0.6	0.1	1620	95	127	4.6	2.1	0.8
2	65	Male	4.9	2.7	190	33	71	7.1	2.9	0.7
3	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0
4	39	Male	1.6	0.8	230	88	74	8.0	4.0	1.0
5	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8
6	73	Male	1.9	0.7	1750	102	141	5.5	2.0	0.5
7	45	Female	1.2	0.5	200	45	50	6.5	3.0	0.75
8	37	Male	0.8	0.3	150	30	40	7.0	3.5	0.85
9	29	Female	0.5	0.1	120	25	30	5.5	2.8	0.9
10	52	Male	2.0	1.0	300	50	60	7.2	3.2	0.95
11	60	Female	1.5	0.6	250	40	55	6.8	3.1	0.8
12	34	Male	0.9	0.2	180	35	45	6.0	2.9	0.88
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3